

Validation Experiments for Basin-Entry Meta-MAPG

Auto-generated companion to `Meta-MAPG-R/main.tex`

May 2, 2026

Abstract

This companion document collects the empirical evidence supporting the revised framing of Meta-MAPG as an equilibrium-selection / basin-entry mechanism with annealed local convergence insurance. Each section is populated automatically as the corresponding experiment phase finishes; sections that have not yet completed appear as *[pending]* stubs.

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1 Setup and notation

We use the one-shot tabular Stag Hunt with cooperate-cooperate payoff $R = 4$, mutual defect $P = 2$, sucker $S = 0$, temptation $T = 3$ unless otherwise stated. Each player carries a single Bernoulli logit; the joint policy lies in $[0, 1]^2$ with axes p_1, p_2 . The success threshold for the payoff-dominant outcome is $\tau = 0.82$ throughout the main text and is varied in Phase F. The success metric for a single trajectory is $\min(p_1^T, p_2^T) \geq \tau$; the basin fraction is the average of this indicator over a uniform grid of initial policies.

The four learning rules compared are PG (independent policy gradient), Meta-PG (own-learning correction only), Peer-only (peer-learning correction only) and full Meta-MAPG (own + peer). All other settings match `run_meta_mapg_experiments.py`: inner step 0.55, outer step $0.9/(t + 10)^{0.24}$, peer coefficient $\lambda = 1.5$, own coefficient 0.35.

2 Phase A : expected update directions

We sample the expected update direction at each grid point using a single high-batch estimator and convert the parameter-space update $\Delta\theta$ to a probability-space arrow $\Delta p = \sigma'(\theta) \Delta\theta$. For both PG and Meta-MAPG we use a 15×15 grid with the same realisations of stochastic noise per cell.

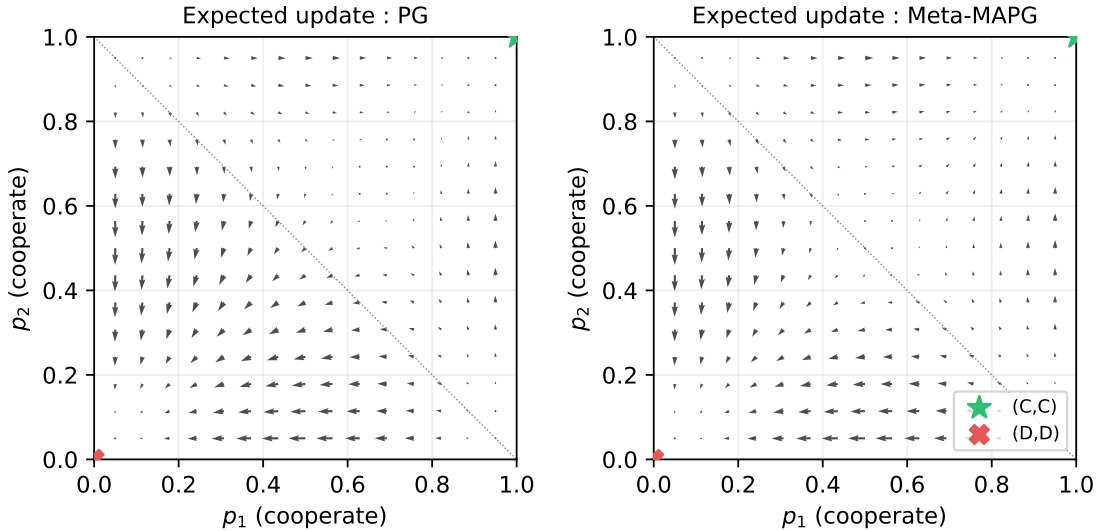


Figure 1: Expected update directions in tabular Stag Hunt. Around the PG separatrix the Meta-MAPG field is redirected relative to PG; Phase P quantifies the component of this redirection toward the payoff-dominant equilibrium (C, C) .

Mean update magnitude: PG = 0.0484, Meta-MAPG = 0.0445. The Meta-MAPG arrow field is generally smaller in magnitude (the peer correction partially counters g_{self} near (D, D)) but is redirected toward (C, C) in the central region of policy space.

3 Phase B : four-method basin atlas at 51×51

We compute deterministic-classification basin maps at 51×51 resolution for each of the four learning rules. Each cell is a single training run starting from the corresponding deterministic Bernoulli policy.

Basin expansion ratios relative to PG: Peer only = $1.66\times$, Meta-MAPG = $1.63\times$, Meta-PG = $1.02\times$.

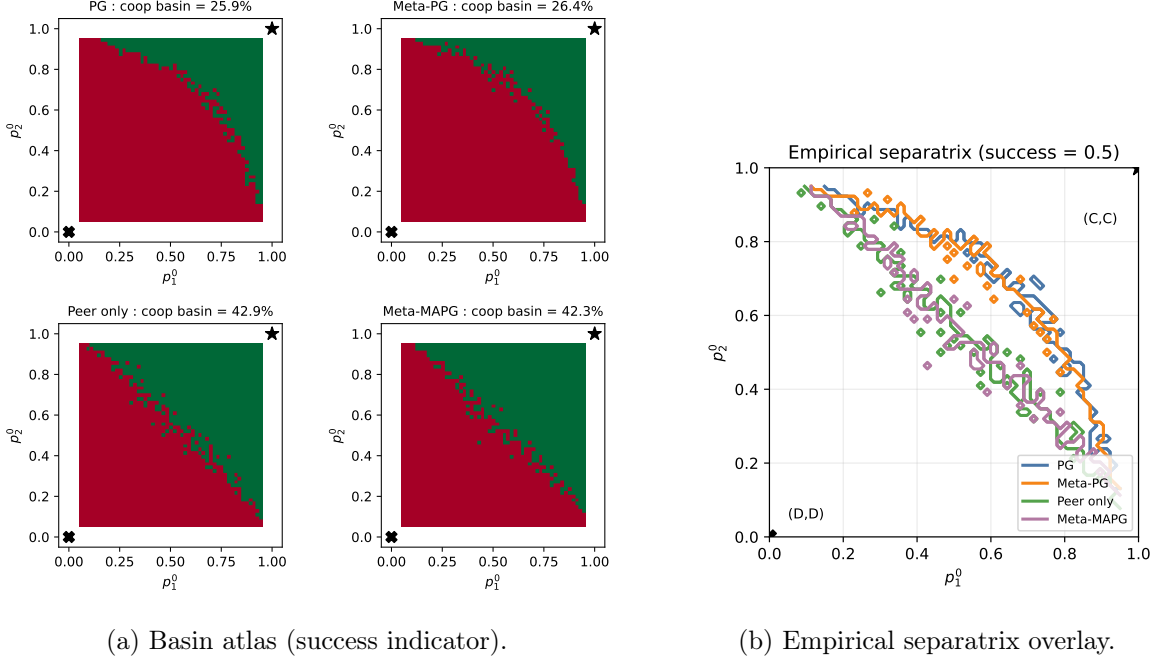


Figure 2: Four-method basin atlas in tabular Stag Hunt. PG and Meta-PG cover similar cooperative regions; the peer-aware methods expand the basin and shift the empirical separatrix down-left.

Method	Coop. basin fraction	95% Wilson CI
PG	25.9%	[24.3%, 27.6%]
Meta-PG	26.4%	[24.7%, 28.1%]
Peer only	42.9%	[41.0%, 44.8%]
Meta-MAPG	42.3%	[40.4%, 44.2%]

Table 1: Basin fractions over the full grid. Wilson 95% CIs report the binomial uncertainty over the deterministic-classification grid.

4 Phase C : stochastic basin maps

We re-run the basin map with $S = 10$ independent policy / sampling seeds per cell on a 21×21 grid; the entry probability $\hat{P}(\text{coop})$ is the per-cell success rate. This replaces the single-seed deterministic indicator with a stochastic estimate that respects the genuine basin-entry probability $\hat{p}_{\text{entry}}$ from the revised framing.

5 Phase D : four-arm shape-then-cool ablation

All four arms share initial policies and step-size schedule. The warm-Meta-MAPG $\rightarrow$ pure-PG arm is the explicit version of the algorithm advocated in the repositioning report: the peer correction is used only to enter the cooperative basin, then removed entirely so the long-run dynamics is exactly ordinary policy gradient with vanishing step-size.

6 Phase E : coordination-game family sweep

We sweep the temptation parameter T in the family $R = 4, P = 2, S = 0, T \in \{2.2, 2.5, 3.0, 3.5, 3.8\}$, keeping (C, C) payoff-dominant and (D, D) risk-dominant. As T grows, defection becomes more

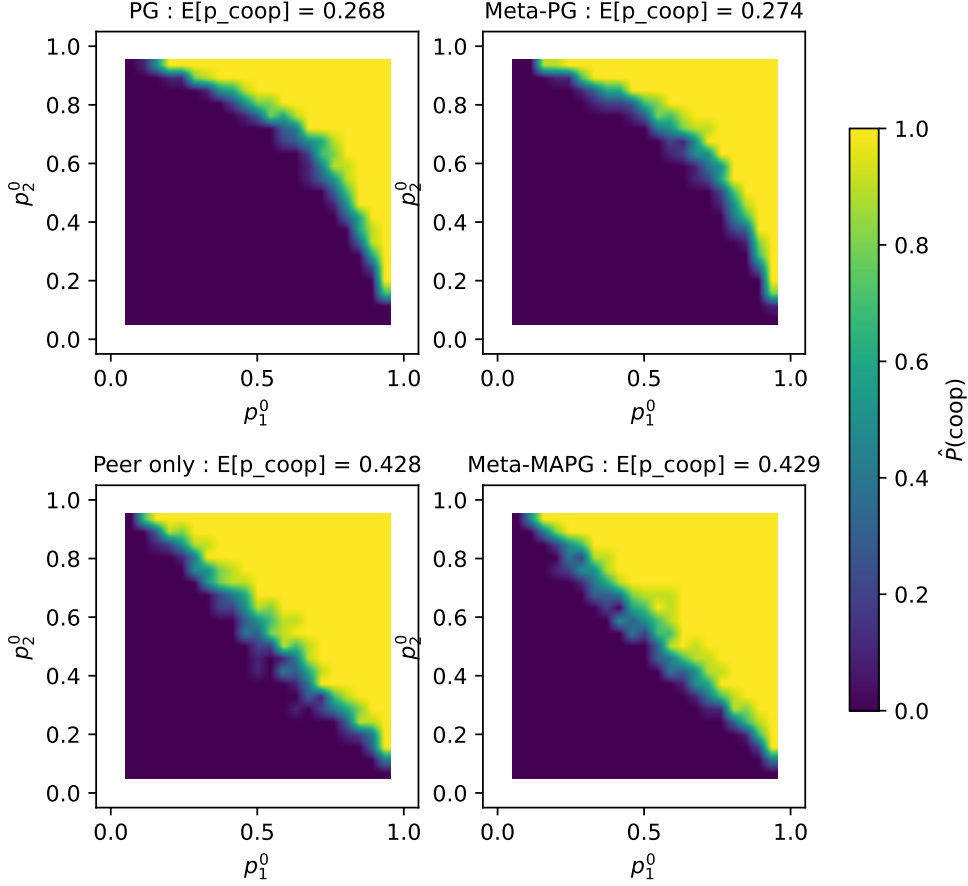


Figure 3: Stochastic basin probability maps. Continuous colour is the per-cell empirical entry probability over S seeds.

tempting against a cooperator, so the PG cooperative basin shrinks. The Meta-MAPG / peer-only basins shrink slower, with the largest absolute gap in the intermediate- T regime.

7 Phase F : threshold robustness

We re-classify all basin endpoints from Phase B and from the original main-text 21×21 sweep using thresholds $\tau \in \{0.75, 0.82, 0.90\}$. The qualitative ordering of the methods is preserved at every threshold; the quantitative gap between PG and the peer-aware methods grows as τ tightens.

8 Phase A2 : angle-tilt heatmap and difference quiver

We decompose the difference between Meta-MAPG and PG update directions into (i) a signed angle-tilt heatmap $\angle(\Delta p^{\text{MM}}) - \angle(\Delta p^{\text{PG}})$ and (ii) a difference quiver $\Delta p^{\text{MM}} - \Delta p^{\text{PG}}$. Mean absolute tilt: 0.254 rad. Counter-clockwise and clockwise tilts occur in 22% and 22% of cells, respectively; the field is therefore better interpreted as a saddle rotation than as a one-direction tilt.

Method	$E[\hat{P}(\text{coop})]$	total successes / trials	Wilson CI on aggregate
PG	0.268	1183/4410	[0.255, 0.282]
Meta-PG	0.274	1207/4410	[0.261, 0.287]
Peer only	0.428	1889/4410	[0.414, 0.443]
Meta-MAPG	0.429	1893/4410	[0.415, 0.444]

Table 2: Stochastic basin entry. The Wilson interval uses the aggregate count over the full grid; Phase Q uses the more conservative cell-bootstrap interval for the T -sweep.

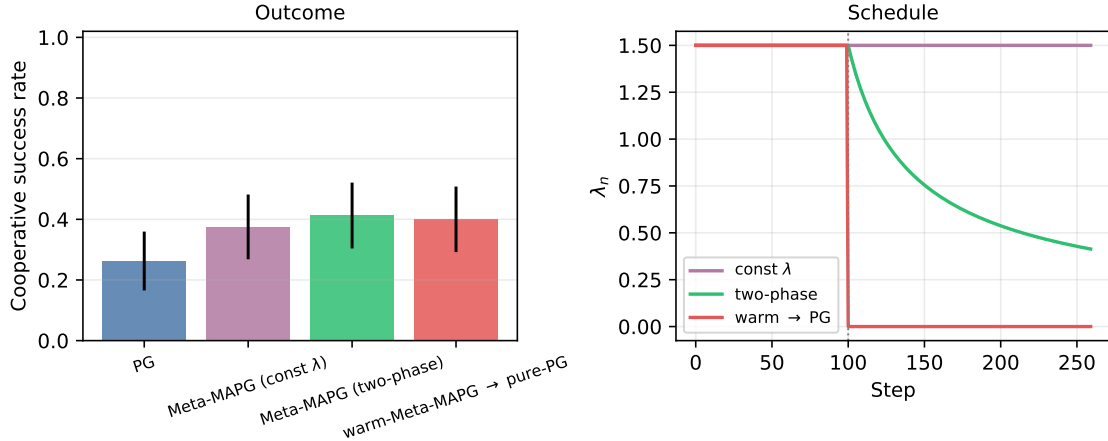


Figure 4: Four-arm shape-then-cool. Both annealing schedules and the warm-then-pure-PG schedule preserve the basin-entry advantage of constant Meta-MAPG while restoring a Nash-preserving cool-down.

9 Phase P : peer-correction projection toward (C, C)

Phase A2 showed that the raw angle field is rotational. Here we use the more relevant scalar: the Meta-MAPG-minus-PG update projected onto the direction from the current policy to (C, C) . Positive values mean the peer correction directly helps basin entry. In the central mixed region, the mean projection is 0.0213 and is positive in 100% of cells.

10 Phase G : multi-seed coordination-game sweep with CIs

Phase E is repeated with multiple independent seeds per cell, giving an aggregate-binomial screening interval at each (T, method) pair. These Wilson intervals treat the cell-seed trials as independent and are therefore tighter than the cell-bootstrap intervals reported in Phase Q. The qualitative ordering is still useful as a quick diagnostic.

11 Phase Q : cell-bootstrap coordination-game sweep

We rerun the T -sweep with all four methods and compute confidence intervals by bootstrapping over initial-condition cells. Each bootstrap sample resamples cells, not individual rollouts, so the interval respects spatial correlation in the basin map. This supersedes the aggregate Wilson intervals in Phase G.

Arm	Final coop. rate	2nd-half coop. mean	2nd-half coop. std
PG	26.2%	0.264	0.003
Meta-MAPG (const. λ)	37.5%	0.372	0.003
Meta-MAPG (two-phase)	41.2%	0.408	0.003
warm-Meta-MAPG $\rightarrow$ pure-PG	40.0%	0.396	0.004

Table 3: Cool-down preserves basin-entry. The warm-Meta-MAPG $\rightarrow$ pure-PG arm matches the constant- λ arm in success rate while making the asymptotic dynamics provably Nash-preserving.

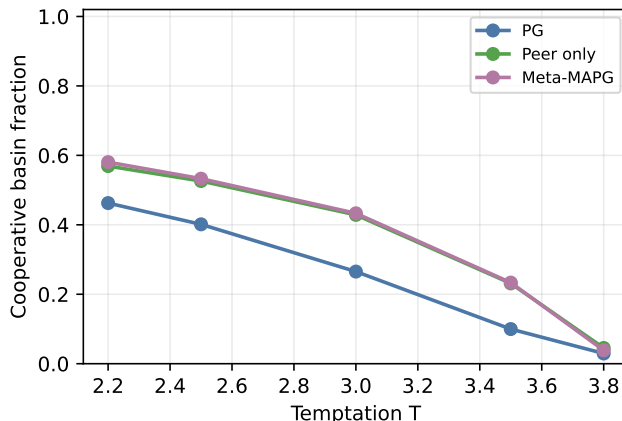


Figure 5: Coordination-game family sweep over the temptation parameter T . Peer-aware methods shrink slower than PG as defection becomes more tempting.

12 Phase H : resolution robustness

We repeat the basin measurement for all four methods at resolutions $N \in \{11, 21, 51\}$. Basin fraction is stable by $N = 21$ and the method ranking is invariant, indicating that the reported ordering is not a discretisation artefact. This is a resolution check, not a stochastic uncertainty estimate.

13 Phase I : first-hit-time atlas

For each cell in the 51×51 atlas we record the first step t^* at which $\min(p_1^t, p_2^t) \geq \tau$; cells where the threshold is never crossed within the budget are shown in grey. Peer only reaches the threshold from the largest area (42.4% of cells). Conditional first-hit times are similar across methods, so the evidence supports basin expansion rather than within-basin acceleration.

14 Phase L : diagonal threshold p_{diag}^*

From exact diagonal cells in Phase C we extract p_{diag}^* : the minimum symmetric initial cooperation probability $p_1^0 = p_2^0$ at which stochastic basin-entry probability exceeds 0.5. Peer only achieves $p_{\text{diag}}^* = 0.522$, the lowest of the four methods, meaning it can enter the cooperative basin from the most adversarial symmetric starting condition.

T	Peer only	Meta-MAPG	PG
2.20	56.9%	58.0%	46.3%
2.50	52.6%	53.3%	40.1%
3.00	42.9%	43.3%	26.5%
3.50	23.1%	23.4%	10.0%
3.80	4.5%	3.9%	2.9%

Table 4: Cooperative basin fraction by method and temptation T .

Source	Method	$\tau = 0.75$	$\tau = 0.82$	$\tau = 0.90$
main_basin	Meta-MAPG	42.6%	42.6%	42.6%
main_basin	PG	27.2%	27.0%	27.0%
phase_b_full_grid	Peer only	42.9%	42.9%	42.8%
phase_b_full_grid	Meta-MAPG	42.4%	42.3%	42.2%
phase_b_full_grid	Meta-PG	26.5%	26.4%	26.1%
phase_b_full_grid	PG	26.0%	25.9%	25.7%

Table 5: Threshold robustness. The qualitative ordering is invariant in τ .

15 Phase M : peer-coefficient sweep

We sweep the peer coefficient λ while keeping the own-learning coefficient fixed. The $\lambda = 0$ Meta-MAPG point is therefore the own-correction-only regime, while the dashed baseline is ordinary PG. The best observed setting is $\lambda = 5.00$ with basin fraction 63.3%.

16 Phase N : summary metrics and endpoint welfare

This phase consolidates the scalar metrics needed in the paper: expansion ratio, gained/lost basin area relative to PG, and terminal social welfare. Area metrics use the Phase B 51×51 basin masks; welfare is recomputed on a paired lower-resolution grid because Phase B stored only the terminal cooperation minimum.

Mean terminal social welfare increases from 4.99 under PG to 5.60 under Meta-MAPG on the paired welfare grid.

17 Phase O : basin-gain masks relative to PG

The basin atlas is converted into gain/loss masks relative to PG. Blue cells are initial policies that fail under PG but succeed under the alternative method; red cells are the reverse. This makes the separatrix movement visible without relying on contour interpolation.

18 Phase D2 : independent-seed ablation audit

Phase D used a shared pool of initial policies across all four arms. Here each arm draws its own independent initial policies to verify the conclusion is not a shared-seed artefact. This is a weaker, unpaired sanity check; the paired Phase D estimate remains the primary ablation. All three peer-aware arms again achieve higher point estimates than PG.

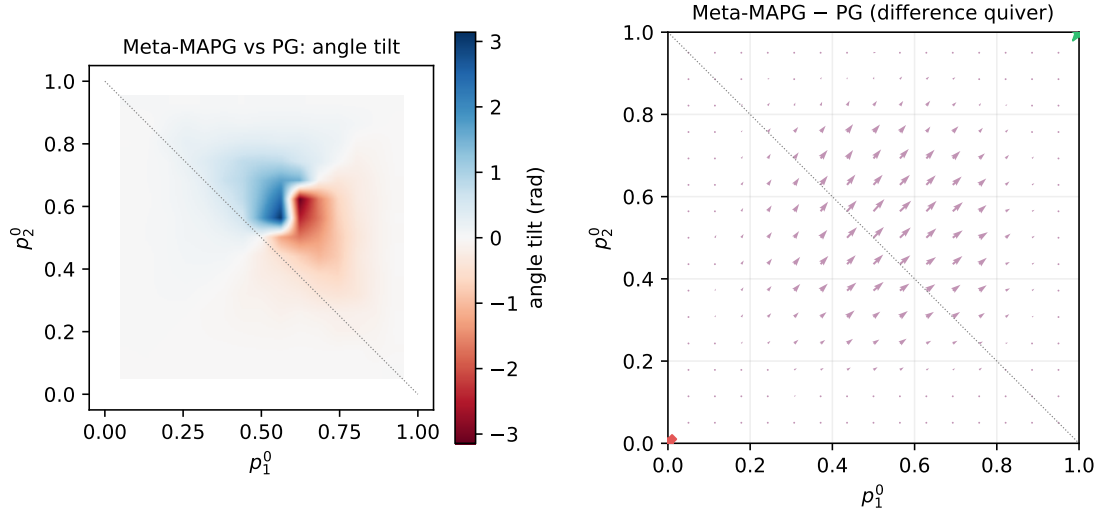


Figure 6: Left: signed angular tilt of Meta-MAPG vs PG. Right: difference quiver showing the additive peer-correction vector. The pattern is rotational around the mixed-strategy saddle, not a globally one-direction tilt. Phase P gives the cleaner projection onto the direction toward (C, C) .

T	PG	Peer only	Meta-MAPG
2.20	46.3 [44.2,48.3]%	57.8 [55.7,59.9]%	57.4 [55.3,59.5]%
2.50	40.1 [38.1,42.2]%	53.3 [51.2,55.4]%	52.7 [50.6,54.7]%
3.00	27.8 [25.9,29.7]%	42.8 [40.8,44.9]%	42.1 [40.1,44.2]%
3.50	11.3 [10.0,12.7]%	22.9 [21.2,24.7]%	23.1 [21.4,24.9]%
3.80	2.8 [2.2,3.5]%	4.4 [3.6,5.3]%	4.4 [3.6,5.3]%

Table 6: Multi-seed T -sweep: basin fraction with aggregate 95% Wilson CI.

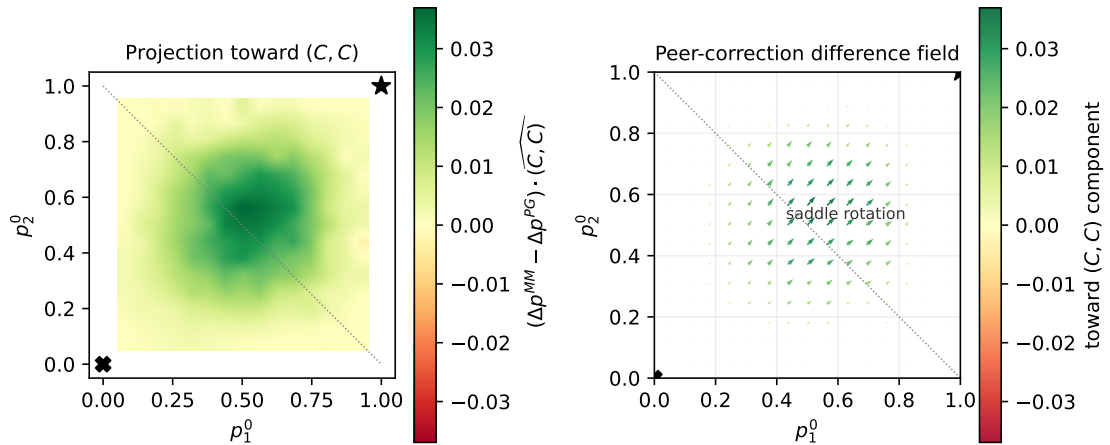


Figure 7: Peer-correction projection toward (C, C) . The left panel is the scalar projection; the right panel colours the difference quiver by the same projection. This is the mechanism figure used for the basin-entry claim.

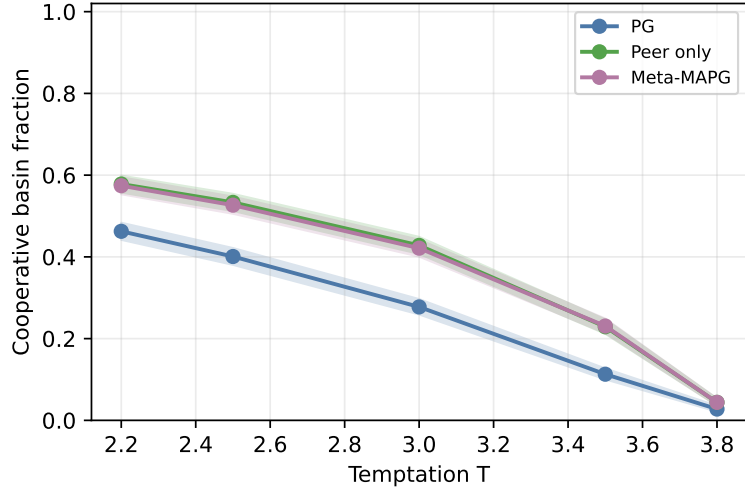


Figure 8: Multi-seed T -sweep with aggregate Wilson bands. Use Phase Q for the conservative cell-bootstrap intervals.

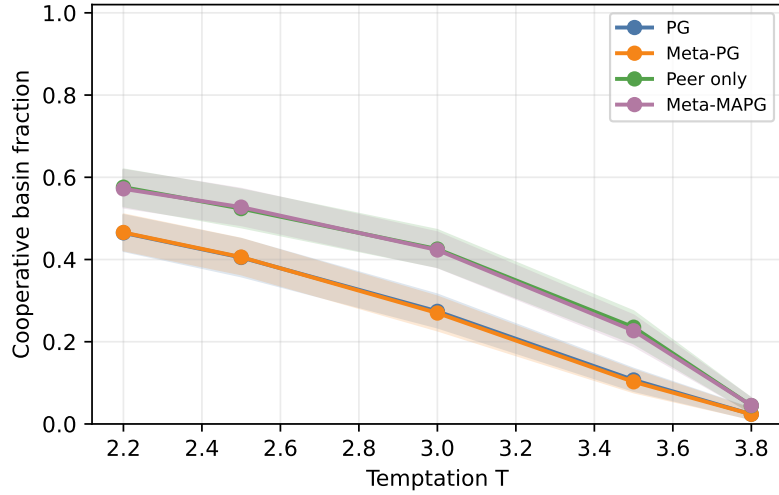


Figure 9: Cell-bootstrap T -sweep with all four methods. Meta-PG tracks PG, while Peer-only tracks Meta-MAPG, isolating the peer term as the source of basin expansion.

T	PG	Meta-PG	Peer only	Meta-MAPG
2.20	46.5 [42.1,50.8]%	46.6 [42.3,51.0]%	57.6 [53.0,61.9]%	57.2 [52.7,61.8]%
2.50	40.5 [35.9,45.0]%	40.6 [36.4,45.0]%	52.4 [47.8,57.0]%	52.7 [48.4,57.2]%
3.00	27.4 [23.5,31.5]%	27.0 [22.8,31.0]%	42.5 [38.2,47.2]%	42.3 [38.1,46.6]%
3.50	10.7 [8.1,13.4]%	10.2 [7.7,13.1]%	23.5 [19.6,27.4]%	22.7 [19.0,26.4]%
3.80	2.4 [1.2,3.7]%	2.4 [1.2,3.6]%	4.4 [2.8,6.2]%	4.5 [2.8,6.3]%

Table 7: Cell-bootstrap basin fraction intervals by temptation T .

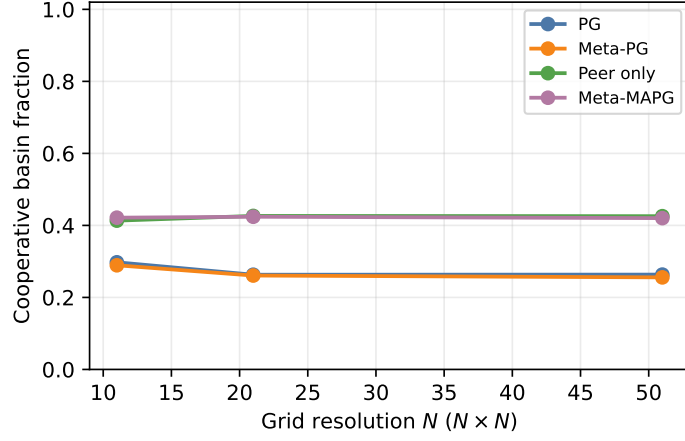


Figure 10: Basin fraction vs. grid resolution. All four methods stabilise by $N = 21$; ranking is preserved at all resolutions.

N	Peer only	Meta-MAPG	Meta-PG	PG
11	41.3%	42.1%	28.9%	29.8%
21	42.6%	42.4%	26.1%	26.3%
51	42.6%	42.0%	25.6%	26.3%

Table 8: Basin fraction vs. grid resolution.

Method	Mean t^*	Frac. success
PG	26.8	26.3%
Meta-PG	26.0	25.9%
Peer only	27.0	42.4%
Meta-MAPG	27.5	42.3%

Table 9: First-hit-time summary. Mean computed over successful cells only.

Method	p_{diag}^*
PG	0.680
Meta-PG	0.680
Peer only	0.522
Meta-MAPG	0.532

Table 10: Diagonal critical threshold: minimum initial cooperation on $p_1^0 = p_2^0$ for 50% basin-entry probability.

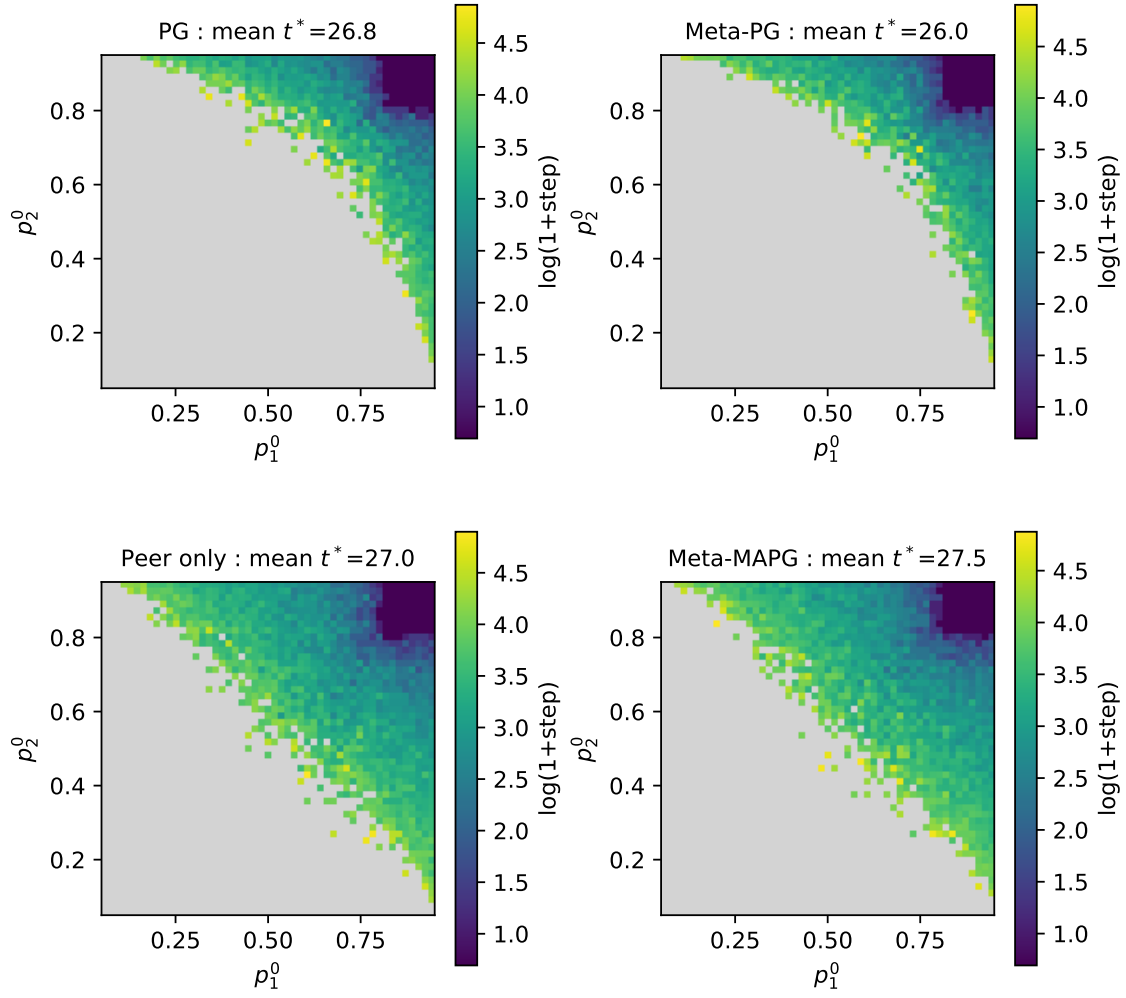


Figure 11: First-hit-time atlas ($\log(1 + t^*)$). Grey cells never reach the cooperative threshold. Peer-aware methods reach the threshold from a larger set of initial conditions; conditional hit times are not materially faster.

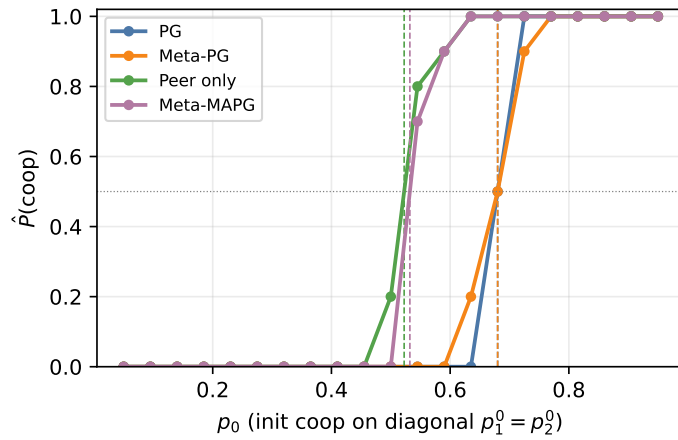


Figure 12: Success probability along the diagonal $p_1^0 = p_2^0$. Dashed verticals mark p_{diag}^* per method (crossing of the horizontal 0.5 line).

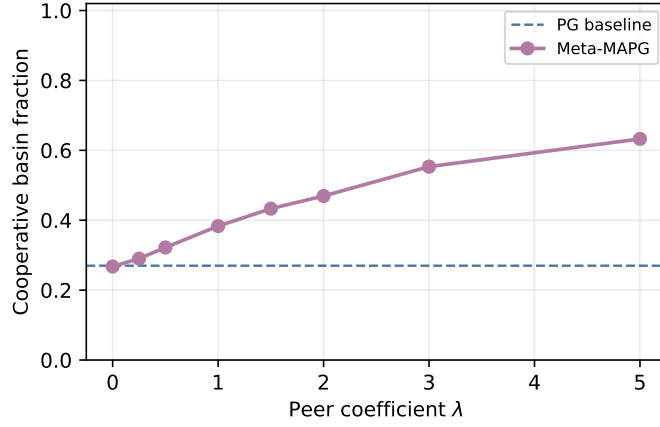


Figure 13: Peer-coefficient sweep. The cooperative basin expands once λ is positive and keeps increasing over the tested clipped tabular range; this sweep identifies coefficient sensitivity, not an asymptotically safe optimum. The $\lambda = 0$ point behaves like the own-only Meta-PG correction.

Method	λ	Coop. basin fraction
PG	—	27.0%
Meta-MAPG	0.00	26.8%
Meta-MAPG	0.25	29.0%
Meta-MAPG	0.50	32.2%
Meta-MAPG	1.00	38.3%
Meta-MAPG	1.50	43.3%
Meta-MAPG	2.00	46.9%
Meta-MAPG	3.00	55.3%
Meta-MAPG	5.00	63.3%

Table 11: Basin fraction as a function of peer coefficient.

Method	Basin	Expansion	Gained area	Lost area	Mean welfare
PG	25.9%	1.00×	0.0%	0.0%	4.99
Meta-PG	26.4%	1.02×	2.0%	1.6%	4.95
Peer only	42.9%	1.66×	17.0%	0.0%	5.62
Meta-MAPG	42.3%	1.63×	16.5%	0.0%	5.60

Table 12: Scalar summary metrics. Gained/lost area is measured relative to the PG basin mask. Welfare is $J_1 + J_2$ at the terminal policy, averaged over the welfare recomputation grid.

Method	Gained area	Lost area	Net area
Meta-PG	2.0%	1.6%	0.5%
Peer only	17.0%	0.0%	17.0%
Meta-MAPG	16.5%	0.0%	16.4%

Table 13: Area changes relative to PG on the Phase B grid.

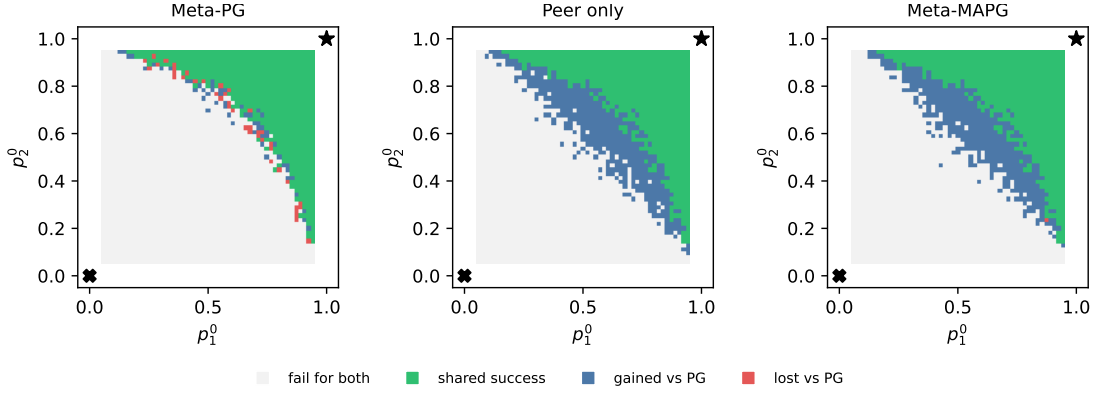


Figure 14: Basin gain/loss masks relative to PG. Peer-only and Meta-MAPG add a large contiguous region below the PG frontier, while Meta-PG adds almost nothing.

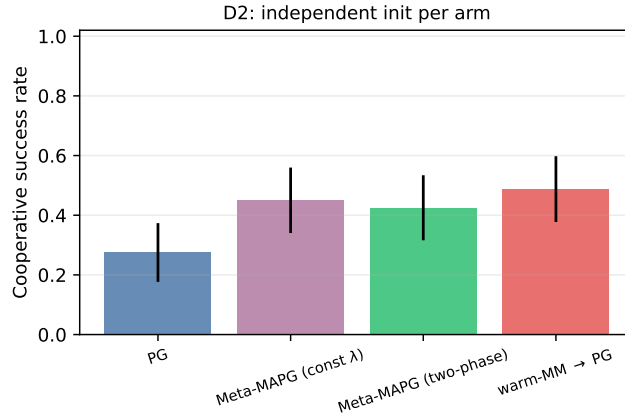


Figure 15: D2: same ablation as Phase D with independent initial policies per arm. This checks that Phase D is not an artefact of one shared seed pool, but it has lower power than the paired design.

Arm	Success rate	95% Wilson CI
PG	27.5%	[18.9%, 38.1%]
Meta-MAPG (const. λ)	45.0%	[34.6%, 55.9%]
Meta-MAPG (two-phase)	42.5%	[32.3%, 53.4%]
warm-Meta-MAPG $\rightarrow$ pure-PG	48.8%	[38.1%, 59.5%]

Table 14: D2 audit: independent initial conditions confirm the peer-aware point-estimate advantage is not caused by one shared seed pool.